Preprints of the 23rd IFAC World Congress
Busan, Republic of Korea, August 23-28, 2026

The Geometry of Hidden Modes in Distance-Based Formation Control ⋆

Solomon Goldgraber Casspi * **Daniel Zelazo** *

* *Stephen B. Klein Faculty of Aerospace Engineering, Technion - Israel Institute of Technology, Haifa, Israel (e-mail: solomon.g@campus.technion.ac.il, dzelazo@technion.ac.il).*

Abstract: This paper presents a geometric input-output analysis of hidden modes in distance-based formation control. We study the linearized dynamics under a gradient control law to characterize the system's structural limitations and their dynamic consequences. Our main contribution is a unified geometric framework for uncontrollable modes. We first prove that uncontrollable rigid-body modes are exactly characterized by a global rotational subspace, $\mathcal{R}_i$. To generalize this, we introduce the local rotational subspace, $\mathcal{T}_i$, which establishes a strict geometric bound on the hidden deformations for minimally connected actuators (i.e., when the actuator node has as many neighbors as the dimension of the space). Finally, we demonstrate the dynamic implications of this structure by proving that the system's ability to recover its shape is determined by an input's alignment with the local component of the standard rotational rigid-body mode, directly linking the geometry of hidden modes to disturbance rejection. We illustrate our results with a case study.

Keywords: Multi-agent systems, Formation control, Control over networks, Controllability.

1. INTRODUCTION

The coordination of multi-agent systems (MASs) is a cornerstone of modern engineering, enabling applications from satellite formation flying to drone swarms and robotic teams (Chung et al. (2018)). A fundamental task in this domain is formation control, where agents must achieve and maintain a specific geometric pattern. In many real-world scenarios, such as indoor or underwater environments, global positioning is unavailable, forcing agents to rely on local, relative sensing. Distance-based formation control, where the desired shape is defined by a set of inter-agent distances, provides a distributed solution for such settings (Oh and Ahn (2011)).

A standard approach to this problem employs a gradient-based control law, where each agent adjusts its motion to minimize the error in its local distance measurements (Krick et al. (2009)). While the stability and convergence properties of these controllers are well-understood (Sun et al. (2016)), their dynamic performance under external disturbances remains an open area of investigation. How does a localized disturbance, such as a wind gust affecting a single drone, propagate through the formation? How does the placement of the actuator and the geometry of the formation itself affect the system's ability to reject such disturbances? To answer these questions, a formal input-output modeling framework is required.

The analysis of controllability in networked systems has a rich history, particularly for linear consensus-type dynamics. Foundational work has established deep connections between a network's controllability and its underlying graph topology, including the role of graph symmetries (Rahmani et al. (2009)) and the placement of leader agents (Tanner (2004)). However, these classical results are often tailored to systems with simple, unweighted graph Laplacians. They do not directly address the challenges posed by distance-based formation control, where the system dynamics are governed by a configuration-dependent, weighted Laplacian that introduces geometric constraints and nonlinearities.

This paper develops a framework to analyze the input-output properties of distance-based formations. We begin by linearizing the nonlinear gradient dynamics around a target equilibrium configuration. This yields a linear time-invariant (LTI) model whose system matrix is a configuration-dependent, weighted graph Laplacian. This model, while an approximation, allows us to leverage the powerful tools of classical systems theory to analyze the formation's controllability.

The main contributions of this paper are twofold. First, we develop a unified geometric framework for uncontrollable modes. We provide an exact geometric characterization of uncontrollable rigid-body modes via the global rotational subspace $\mathcal{R}_i$, and establish a geometric bound on uncontrollable deformations via the local rotational subspace $\mathcal{T}_i$ for minimally connected actuators. Second, we analyze the dynamic implications of this structure, demonstrating that the system's ability to recover its shape is determined by the alignment of a disturbance with the local component of the standard rotational rigid-body mode. This establishes a direct link between the geometry of hidden modes and disturbance rejection.

⋆ This work was supported by the Israel Science Foundation grant no. 453/24 and the Gordon Center for Systems Engineering.

©2026 the authors. Accepted by IFAC for publication under a Creative Commons Licence CC-BY-NC-ND

The remainder of this paper is organized as follows. Section 2 provides the necessary background on rigidity theory and system-theoretic concepts. Section 3 formally defines the problem, introducing the nonlinear dynamics and the linearized model used for analysis. Section 4 presents our main theoretical contribution: a geometric characterization of the uncontrollable subspace. Finally, Section 5 analyzes the dynamic implications of this geometric structure and illustrates the results with a case study.

Notations Throughout, I_d denotes the identity matrix in $\mathbb{R}^d$, e_k is the k-th standard basis vector in $\mathbb{R}^n$, and Ω denotes a $d \times d$ skew-symmetric matrix representing an infinitesimal rotation. For $d = 2$, this is uniquely represented by $\Omega = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. The eigenspace associated with an eigenvalue λ is denoted by E_λ. The Kronecker product is denoted by $\otimes$. For a stacked vector $v \in \mathbb{R}^{nd}$, its component corresponding to agent i is denoted $v_i \in \mathbb{R}^d$. The degree of a node i in $\mathcal{G}$, i.e., the number of edges incident to i, is denoted $\deg(i)$. The standard Euclidean inner product is denoted by $\langle \cdot, \cdot \rangle$.

2. PRELIMINARIES

This section provides the necessary background on graph rigidity and system-theoretic concepts.

2.1 Rigidity of Graphs

A formation of n agents is modeled as a *framework* $(\mathcal{G}, p)$, where $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is an undirected graph with a vertex set $\mathcal{V} = \{1, \dots, n\}$ and an edge set $\mathcal{E}$ of size $m = |\mathcal{E}|$. The vector $p \in \mathbb{R}^{nd}$ is the stacked vector of agent positions. A reference configuration, used to define the desired inter-agent distances, is denoted p^*.

The shape of the framework is encoded by the vector of squared inter-agent distances, known as the *rigidity function*, $r(p) : \mathbb{R}^{nd} \to \mathbb{R}^m$, with

$$r_k(p) = \tfrac{1}{2}\|p_i - p_j\|^2, \quad \text{for each } k = (i, j) \in \mathcal{E}. \tag{1}$$

The first-order change in these distances due to an infinitesimal motion $v \in \mathbb{R}^{nd}$ about the reference configuration p^* is given by the Taylor expansion $r(p^* + tv) - r(p^*) = tR(p^*)v + o(t^2)$, where the Jacobian $R(p^*) := \nabla_p r(p)|_{p=p^*} \in \mathbb{R}^{m \times nd}$ is the *rigidity matrix* evaluated at p^* (Asimow and Roth (1978, 1979)).

Any distance-preserving first order motion must reside in the nullspace $\ker(R(p^*))$. These allowable motions fall into two distinct physical categories: *rigid body motions* (RBMs), representing trivial translations and rotations of the entire configuration, and *infinitesimal flexes*, representing internal shape deformations. A framework is infinitesimally rigid if its nullspace consists entirely of RBMs, completely lacking any internal flexes (Connelly (2005)). For a generic framework in $\mathbb{R}^2$, this is equivalent to the rank condition $\operatorname{rank}(R(p^*)) = 2n - 3$ (Laman (1970)).

The rigidity matrix also defines two other fundamental subspaces. The nullspace of its transpose, $\ker(R(p^*)^\top)$, is the space of *self-stresses*. A non-zero self-stress is a set of tensions and compressions on the edges that self-equilibrate at every node. By the fundamental theorem of linear algebra, the image of the transpose, $\operatorname{Im}(R(p^*)^\top)$, is the orthogonal complement to the space of RBMs. It therefore represents the space of all infinitesimal *deformations*.

2.2 Controllability of LTI Systems

Consider a general LTI system $\dot{x} = Ax + Bu$. The system's controllability properties determine which internal modes can be influenced by the input. A key tool for this analysis is the Popov-Belevitch-Hautus (PBH) test: a mode represented by an eigenvector v of A is *uncontrollable* if and only if it lies in the nullspace of $B^\top$, i.e., $B^\top v = 0$. The set of all such modes spans the uncontrollable subspace $\overline{\mathcal{C}}$, which we refer to as the *hidden* subspace of the system (Skogestad and Postlethwaite (2005)).

3. PROBLEM FORMULATION

We consider a formation of n agents in $\mathbb{R}^d$ (typically $d = 2$) with single-integrator dynamics, $\dot{p}_i = u_i$, a standard model in the distance-based formation control literature (Krick et al. (2009); Oh and Ahn (2011); Babazadeh and Selmic (2019)). The agent interactions are described by an undirected sensing graph $\mathcal{G}$.

The control objective is to steer the formation to a configuration p that satisfies a desired set of inter-agent distances, defined by a reference configuration p^*. This is achieved using a gradient-descent control law. We define a potential function based on the collective error in squared edge lengths,

$$V(p) = \frac{1}{2}\|r(p) - r(p^*)\|^2, \tag{2}$$

where $r(p)$ is the rigidity function defined (1). The control law is the negative gradient of this potential, $\dot{p} = -\nabla_p V(p)$, which is a standard approach in the literature (see, e.g., Krick et al. (2009); Oh et al. (2015)). This yields the closed-loop dynamics:

$$\dot{p} = -R(p)^\top \big(r(p) - r(p^*)\big). \tag{3}$$

Note that V is invariant under continuous rigid-body motions of the formation. Consequently, the equilibrium condition $r(p) = r(p^*)$ does not define an isolated point, but a continuous family of translated and rotated copies of the reference configuration.

To analyze the system's response to a localized exogenous disturbance, we augment these dynamics with a single-channel input applied at one agent i,

$$\dot{p} = -R(p)^\top \big(r(p) - r(p^*)\big) + Bw, \tag{4}$$

where $w \in \mathbb{R}^d$ and $B = e_i \otimes I_d$. The single-actuator setting is the simplest abstraction in which to expose how actuator placement and graph geometry interact, and serves as a building block for multi-input extensions.

Direct analysis of the nonlinear model (4) is challenging. To gain tractable insights, we linearize the system around the equilibrium point $(p^*, w = 0)$. Letting $\delta p = p - p^*$, the first-order Taylor expansion of the dynamics yields the LTI model:

$$\delta\dot{p} = A\,\delta p + B\,\delta w, \tag{5}$$

where the system matrix A is the weighted graph Laplacian, or "stiffness matrix," given by:

$$A = -R(p^*)^\top R(p^*). \tag{6}$$

The analysis that follows characterizes the input-output behavior of (5) in a neighborhood of p^* where this linearization is valid; all subsequent statements, including the shape-recovery dichotomy of Section 5, are properties of this linearized model. A geometric treatment of the full nonlinear dynamics is the subject of ongoing work.

4. GEOMETRIC CHARACTERIZATION OF HIDDEN MODES

This section develops a geometric characterization of the hidden modes for the linearized system (5). We first establish their existence and algebraic structure before revealing their geometric form.

4.1 Algebraic Foundations of Hidden Modes

We begin by establishing that for an infinitesimally rigid formation, hidden modes are an unavoidable consequence of using a single actuator.

Proposition 1. (Existence of Hidden RBMs). Consider the linearized system (5) for an infinitesimally rigid framework in $\mathbb{R}^d$ with $d \geq 2$, actuated at a single node i. Let $\overline{\mathcal{C}}$ be the uncontrollable subspace and $E_0 = \ker(R(p^*))$ be the *rigid body motion* (RBM) eigenspace. Then the dimension of the uncontrollable RBM subspace satisfies

$$\dim(\overline{\mathcal{C}} \cap E_0) \geq \frac{d(d-1)}{2} \geq 1.$$

For any $d \geq 2$, this guarantees the existence of at least one uncontrollable RBM.

Proof. The RBM eigenspace E_0 corresponds to the zero-eigenspace of the system matrix $A = -R(p^*)^\top R(p^*)$. For an infinitesimally rigid framework in $\mathbb{R}^d$, this space has dimension $\dim(E_0) = d(d+1)/2$. By the PBH test, the uncontrollable subspace is $\overline{\mathcal{C}} = \bigoplus_\lambda (E_\lambda \cap \ker(B^\top))$. The uncontrollable part of the RBM eigenspace is therefore $E_0 \cap \ker(B^\top)$. The dimension of this subspace is given by:

$$\begin{aligned}\dim(E_0 \cap \ker(B^\top)) &\geq \dim(E_0) - \operatorname{rank}(B) \\ &= \frac{d(d+1)}{2} - d = \frac{d(d-1)}{2}.\end{aligned}$$

For any $d \geq 2$, we have $d(d-1)/2 \geq 1$, which proves the existence of at least one uncontrollable RBM. □

Having established their existence, we now seek to understand their structure.

Proposition 2. (Uncontrollable Subspace Decomposition). The total uncontrollable subspace $\overline{\mathcal{C}}$ can be decomposed into its RBM and deformational components as:

$$\overline{\mathcal{C}} = \big(\ker(R(p^*)) \cap \ker(B^\top)\big) \oplus \big(\operatorname{Im}(R(p^*)^\top) \cap \ker(B^\top)\big), \tag{7}$$

where $\ker(R(p^*))$ is the RBM subspace and $\operatorname{Im}(R(p^*)^\top)$ is the subspace of shape-changing deformations.

Proof. The decomposition follows from the orthogonal decomposition of the symmetric matrix $A = -R(p^*)^\top R(p^*)$ into $E_0 = \ker(R(p^*))$ and $\bigoplus_{\lambda \neq 0} E_\lambda = \operatorname{Im}(R(p^*)^\top)$, after intersecting each component with $\ker(B^\top)$. □

This decomposition provides the algebraic foundation. To find a geometric interpretation for each term in (7), we apply the PBH test to our single-actuator system. With $B = e_i \otimes I_d$, the uncontrollability condition $B^\top v = 0$ simplifies to a direct "pinning" constraint at the agent level.

Lemma 3. (Pinning Constraint). An eigenvector v is uncontrollable from an input applied at agent i if and only if its component at that agent vanishes, i.e., $v_i = 0$.

Proof. From the PBH test, v is uncontrollable iff $B^\top v = (e_i^\top \otimes I_d)v = v_i = 0$. □

4.2 Geometric Characterization of Hidden RBMs

We now use the pinning constraint formulated in Lemma 3 to find a geometric interpretation for the RBM and deformational components of the uncontrollable subspace. We first ask: what is the geometric form of the guaranteed hidden RBM from Proposition 1? Applying the pinning constraint to a general RBM reveals its structure.

Proposition 4. (Uncontrollable RBM is a Pure Rotation). The uncontrollable RBM subspace, $\ker(R(p^*)) \cap \ker(B^\top)$, is the $d(d-1)/2$-dimensional rotational subspace $\mathcal{R}_i(p^*)$. This subspace consists of all infinitesimal motions corresponding to a pure rotation of the framework about the actuator node i,

$$\mathcal{R}_i(p^*) := \left\{ v \in \mathbb{R}^{dn} \mid v_j = \Omega(p_j^* - p_i^*),\ \forall j \in \mathcal{V} \right\}.$$

Proof. Any RBM admits a unique decomposition $v_j = b + \Omega(p_j^* - p_{\mathrm{cm}})$ into a translation b and a rotation about the centroid. The pinning constraint $v_i = 0$ from Lemma 3 fixes $b = -\Omega(p_i^* - p_{\mathrm{cm}})$, which after substitution yields $v_j = \Omega(p_j^* - p_i^*)$, a pure rotation about node i. □

This geometric interpretation is illustrated in Figure 1. This proof reveals a key geometric insight: the algebraic "pinning constraint" $v_i = 0$ forces the center of rotation of any hidden RBM to coincide with the actuated node i. This establishes a direct and intuitive link between the placement of the actuator and the geometric form of the resulting uncontrollable mode.

4.3 Locally Hidden Deformations

Having characterized the hidden RBMs, we now generalize our approach to find hidden deformations. The gradient controller at node i is driven only by errors in its incident edges. A motion is therefore "locally hidden" from the controller at node i if it generates no first-order change in these local edge lengths.

To formalize this, we consider the components of the rigidity function, $r_k(p) = \frac{1}{2}\|p_i - p_j\|^2$, corresponding to the edges incident to node i. These form the *star relations* at i. The first-order change in these relations due to an infinitesimal motion v is given by

$$\delta r_k(p) = (p_i^* - p_j^*)^\top (v_i - v_j), \text{ for each edge } k = (i,j) \in \mathcal{E}.$$

A motion v is locally hidden from node i if it produces no change in these local measurements, i.e., $\delta r_k(p) = 0$ for all edges j incident to i. This motivates the following definition, which adds the pinning constraint $v_i = 0$.

Definition 5. (The Local Rotational Subspace $\mathcal{T}_i$). The set of infinitesimal motions that fix node i and preserve the

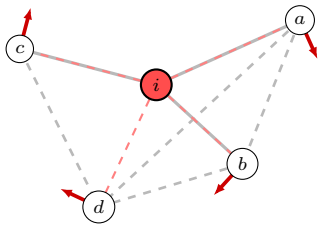


Fig. 1. Geometric interpretation of the uncontrollable rotational subspace $\mathcal{R}_i(p^*)$. The red arrows depict a pure rotation of the *entire* framework about the actuated node i (red filled). Dashed red lines indicate the rotation arms from i to every other node, showing that this global rigid-body motion is generated jointly by all relative positions with respect to i. The dashed gray edges show the underlying sensing graph.

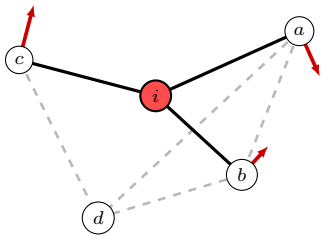


Fig. 2. The local rotational subspace $\mathcal{T}_i$ at the actuated node i (red). Solid black edges form the incident star; dashed gray edges are non-incident. The red arrows depict elementary motions $\tau_{i\to j}$, where each neighbor $j \in \{a, b, c\}$ can rotate independently about i. Because these local rotations are decoupled, they generate internal shape deformations along the non-incident edges. These local motions only manifest as true dynamic hidden modes if they simultaneously satisfy the global eigenvector conditions of $\overline{\mathcal{C}}$.

lengths of all its incident edges to first order, denoted $\mathcal{T}_i(\mathcal{G}, p^*)$, is defined as

$$\mathcal{T}_i := \{v \in \mathbb{R}^{dn} : v_i = 0 \text{ and } (p_j^* - p_i^*)^\top v_j = 0, \forall (i,j) \in \mathcal{E}\}.$$

To compute the dimension of $\mathcal{T}_i$, we construct an explicit basis by counting the constraints in Definition 5. The pinning condition $v_i = 0$ removes d degrees of freedom from the ambient $\mathbb{R}^{dn}$. The remaining $d(n-1)$ dimensions split according to the actuator's neighborhood:

(i) For each neighbor j of i, the length-preserving condition $(p_j^* - p_i^*)^\top v_j = 0$ removes one additional degree of freedom, confining v_j to the $(d-1)$-dimensional hyperplane orthogonal to the incident edge. In the planar case ($d = 2$), this hyperplane reduces to the line spanned by the elementary rotation

$$\tau_{i\to j} := (e_j \otimes I_d)\,\Omega\,(p_j^* - p_i^*), \tag{8}$$

which rotates only neighbor j infinitesimally about i.

(ii) Each non-neighbor ℓ is unconstrained by Definition 5 and contributes its full d standard basis directions.

Summing these contributions yields

$$\dim(\mathcal{T}_i) = d(n-1) - \deg(i).$$

By construction, every motion in $\mathcal{T}_i$ preserves the edges incident to i and is therefore locally hidden to the gradient law at i. At the same time, such motions are not globally rigid: a neighbor j rotating about i leaves the edge (i,j) intact but generally changes all distances from j to other nodes, producing a shape-changing deformation that goes undetected by the actuator. These elementary motions are illustrated in Figure 2.

Proposition 6. (Geometric Subspace Inclusion). The rotational subspace $\mathcal{R}_i(p^*)$ is a subspace of the local rotational subspace $\mathcal{T}_i(\mathcal{G}, p^*)$, i.e., $\mathcal{R}_i(p^*) \subseteq \mathcal{T}_i(\mathcal{G}, p^*)$.

Proof. For $v \in \mathcal{R}_i(p^*)$, $v_i = \Omega(p_i^* - p_i^*) = 0$ satisfies the pinning constraint, and $(p_j^* - p_i^*)^\top \Omega (p_j^* - p_i^*) = 0$ by skew-symmetry of Ω. □

The inclusion highlights a key distinction: $\mathcal{R}_i$ contains only the single global RBM that is a pure rotation about i, whereas $\mathcal{T}_i$ also contains the elementary motions $\tau_{i\to j}$, which are locally rigid at i but globally deformational and lie in $\mathrm{Im}(R(p^*)^\top)$.

Theorem 7. (Geometric Bound for $\deg(i) = d$). For an infinitesimally rigid framework $(\mathcal{G}, p^*)$ actuated at a node i with minimal degree $\deg(i) = d$, the uncontrollable subspace is geometrically bounded by the local rotational subspace, i.e.,

$$\overline{\mathcal{C}} \subseteq \mathcal{T}_i(\mathcal{G}, p^*).$$

Proof. Let $v \in \overline{\mathcal{C}}$ be an uncontrollable mode. By the PBH test and Lemma 3, v must satisfy the pinning constraint $v_i = 0$ and the eigenvector equation $-R(p^*)^\top R(p^*)v = \lambda v$.

Evaluating the i-th vector block of the eigenvector equation yields the equilibrium condition at the actuator:

$$\sum_{j \in \mathcal{N}_i} (p_i^* - p_j^*)\delta r_{ij} = 0, \tag{9}$$

where $\delta r_{ij} = -(p_i^* - p_j^*)^\top v_j$ represents the first-order extension of the incident edge (i,j) given $v_i = 0$.

Since the framework is infinitesimally rigid, the rank condition of the rigidity matrix inherently requires $\deg(i) \geq d$ to prevent local infinitesimal flexes. By our premise, $\deg(i) = d$. For generic configurations, these d incident edge vectors form a basis for $\mathbb{R}^d$ and are therefore linearly independent. The equilibrium condition (9) is satisfied if and only if $\delta r_{ij} = 0$ for all $j \in \mathcal{N}_i$.

Consequently, the mode v satisfies both the pinning constraint ($v_i = 0$) and the preservation of all incident edges ($\delta r_{ij} = 0$). These are exactly the defining geometric conditions of Definition 5, implying $v \in \mathcal{T}_i$. Therefore, the strict geometric inclusion $\overline{\mathcal{C}} \subseteq \mathcal{T}_i$ holds. □

By substituting the geometric bound from Theorem 7 into the algebraic decomposition of Proposition 2, and

accounting for the rigid-body inclusion ($\mathcal{R}_i \subset \mathcal{T}_i$) from Proposition 6, we immediately isolate the deformational hidden modes.

Corollary 8. For an infinitesimally rigid framework actuated at a node i with $\deg(i) = d$, the algebraic decomposition of the uncontrollable subspace from Proposition 2 is geometrically bounded by:

$$\overline{\mathcal{C}} \subseteq \mathcal{R}_i(p^*) \oplus \big(\mathcal{T}_i(\mathcal{G}, p^*) \cap \mathrm{Im}(R(p^*)^\top)\big).$$

Remark 9. (Dimension of Hidden Deformations). Combining the dimension of $\mathcal{R}_i$ from Section 4.2 with the dimension count of Section 4.3, and using the orthogonal decomposition in Corollary 8, gives a closed form for the size of this bounding subspace. Since $\dim(\mathcal{R}_i) = d(d-1)/2$ and $\dim(\mathcal{T}_i) = d(n-1) - \deg(i)$, orthogonality yields

$$\begin{aligned}\dim\big(\mathcal{T}_i(\mathcal{G}, p^*) \cap \mathrm{Im}(R(p^*)^\top)\big) &= dn - \tfrac{d(d+1)}{2} - \deg(i) \\ &= \dim(\mathrm{Im}(R(p^*)^\top)) - \deg(i).\end{aligned}$$

While increasing $\deg(i)$ strictly reduces the dimension of this bounding subspace, pushing $\deg(i) > d$ introduces linearly dependent edges that theoretically permit hidden modes outside $\mathcal{T}_i$ via equilibrium stresses. However, because all deformational modes in this LTI system are exponentially stable transients, minimizing this bounding subspace via highly connected actuators remains a practical structural design heuristic, as the global network dynamics generally suppress these over-constrained modes.

5. DYNAMIC IMPLICATIONS OF HIDDEN MODES

We now analyze the dynamic consequences of the hidden modes, investigating how a localized input at agent i affects the formation's ability to recover its shape. This reveals a direct link between the geometry of the uncontrollable subspace $\mathcal{R}_i$ and the system's steady-state response.

We analyze the system's steady-state response to an impulsive input, $w(t) = w_0\delta(t)$, where $w_0 \in \mathbb{R}^d$ is the direction of the impulse. The final state deviation is given by the projection of the initial impulse onto the RBM eigenspace $E_0 = \ker(R(p^*))$,

$$\lim_{t\to\infty} \delta p(t) = P_0 B w_0, \tag{10}$$

where P_0 is the orthogonal projection matrix onto E_0. Let $\{v_x, v_y, v_r\}$ be an orthonormal basis for the RBM space, representing x-translation, y-translation, and rotation about the center of mass, respectively. The final state will be a linear combination of these modes,

$$\lim_{t\to\infty} \delta p(t) = c_x v_x + c_y v_y + c_r v_r, \tag{11}$$

where the coefficients are determined by the input:

$$c_x = \langle v_x, Bw_0\rangle,\ c_y = \langle v_y, Bw_0\rangle,\ c_r = \langle v_r, Bw_0\rangle.$$

The key insight is that shape recovery depends entirely on whether the global rotational mode v_r is excited (i.e., if $c_r \neq 0$). The coefficient of the rotational mode in the final state is given by the inner product $c_r = \langle v_r, Bw_0\rangle = \langle [v_r]_i, w_0\rangle$, where $[v_r]_i$ is the local velocity of the rotational mode at the actuated node, a defining component of the uncontrollable rotational subspace $\mathcal{R}_i$.

Proposition 10. (Shape Recovery). Consider an impulsive input $w(t) = w_0\delta(t)$ applied at agent i. The ability of the formation to recover its shape is determined by the alignment between the input vector w_0 and the local rotational vector $[v_r]_i$. This vector is a defining component of the uncontrollable rotational subspace $\mathcal{R}_i$.

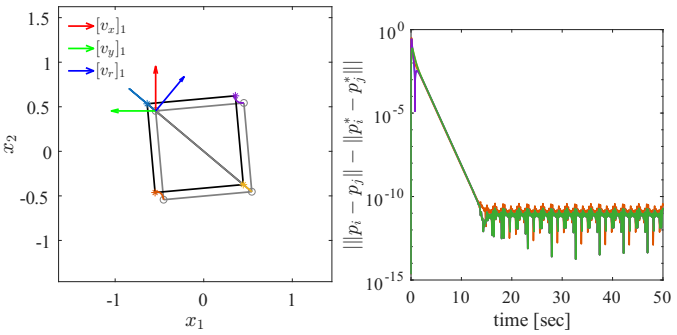
(i) *Perfect Shape Recovery:*
If the input is orthogonal to the local rotational direction, $\langle [v_r]_i, w_0\rangle = 0$, then the rotational mode is not excited ($c_r = 0$). The resulting steady-state motion is a pure translation of the entire framework. All inter-agent distances are preserved, and the steady-state edge error is zero.

(ii) *Persistent Shape Distortion:*
If the input has a component along the local rotational direction, $\langle [v_r]_i, w_0\rangle \neq 0$, then the rotational mode is excited ($c_r \neq 0$). The resulting steady-state motion is a combination of translation and rotation, which causes a persistent change in the inter-agent distances and a non-zero steady-state edge error.

Proof. The coefficient is $c_r = \langle v_r, (e_i \otimes I_d)w_0\rangle = \langle [v_r]_i, w_0\rangle$. If $c_r = 0$, the steady-state motion is a pure translation, $[\delta p(\infty)]_k = b$, which preserves all relative positions $p_j^* - p_l^*$ and hence all edge lengths. If $c_r \neq 0$, an edge (j,l) becomes $(I_d + c_r\Omega)(p_j^* - p_l^*)$; using $\Omega^\top = -\Omega$, its squared length is $\|x\|^2 + c_r^2\|\Omega x\|^2 > \|x\|^2$ for any non-zero edge vector $x = p_j^* - p_l^*$, so the edge error is persistent. □

This result reveals a fundamental dichotomy in the system's response, illustrated in Figures 3 and 4, and discussed in detail in the following subsection.

5.1 Case Study: Minimally Rigid Framework

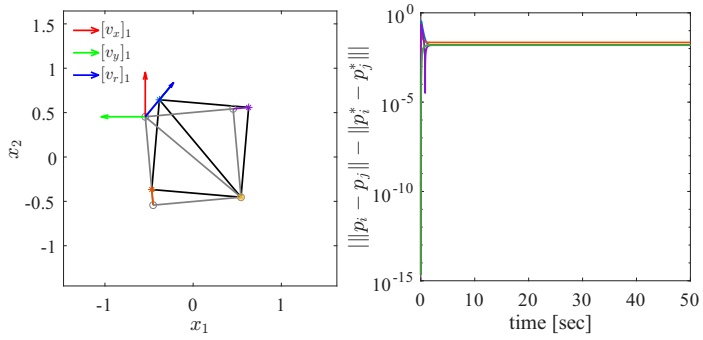


(a) (b)

Fig. 3. Shape recovery with an orthogonal input. (a) An impulse orthogonal to the local rotational RBM direction at the actuated node results in a pure translation. (b) The edge length errors decay to zero.

To illustrate the shape recovery dichotomy, we consider a minimally rigid formation of four agents in the plane, as shown in Figures 3 and 4. An impulsive input is applied at node 1.

First, we consider an input w_0 orthogonal to the local rotational vector at the actuated node, $\langle [v_r]_1, w_0\rangle = 0$. This corresponds to Case 1 of Proposition 10, which predicts that the rotational mode is not excited ($c_r = 0$). Figure 3(a) shows the resulting motion: the formation undergoes a pure translation. Figure 3(b) confirms this, showing that all edge length errors decay to zero after the



(a) (b)

Fig. 4. Shape distortion with an aligned input. (a) An impulse with a component along the local rotational RBM direction excites a rotational motion. (b) The formation deforms, resulting in persistent edge length errors.

initial transient, and the formation perfectly recovers its shape.

Next, we consider an input w_0 with a component along the local rotational vector, $\langle [v_r]_1, w_0 \rangle \neq 0$ (Case 2). This input excites the rotational mode ($c_r \neq 0$): Figure 4(a) shows that the final motion combines translation with a rigid rotation, and Figure 4(b) shows the edge length errors converge to persistent, non-zero values, providing a clear dynamic validation of our geometric analysis.

The dichotomy admits a clean geometric interpretation in RBM coordinate space. By Lemma 3, the uncontrollable direction in the three-dimensional RBM coordinate space (c_x, c_y, c_r) is $n_c = \begin{pmatrix} -[v_r]_i \\ 1 \end{pmatrix}$; every input w_0 produces reachable RBM coordinates lying in the orthogonal "controllable plane" shown in Figure 5. Inputs orthogonal to $[v_r]_i$ further restrict $c_r = 0$, placing the outcome on the intersection of this plane with the translation subspace and yielding perfect shape recovery.

6. CONCLUSION

We presented a geometric input-output analysis of hidden modes in distance-based formation control under a single localized actuator. For minimally connected actuators ($\deg(i) = d$), we proved that the uncontrollable subspace is geometrically bounded by the local rotational subspace $\mathcal{T}_i$. This yields an orthogonal decomposition isolating the exact global rotational subspace $\mathcal{R}_i$ on the rigid-body component, and a bounded deformational complement $\mathcal{T}_i \cap \mathrm{Im}(R(p^*)^\top)$. Because all deformational hidden modes in this LTI system are exponentially stable transients, the formation's steady-state response is restricted entirely to the rigid-body eigenspace. This enabled us to prove a shape-recovery dichotomy from a purely local input-output perspective: a localized impulsive disturbance produces persistent shape distortion if and only if its component along the local rotational RBM $[v_r]_i$ is non-zero. Future work will investigate the exact geometric characterization of over-actuated nodes ($\deg(i) > d$), where local equilibrium stresses theoretically permit hidden modes outside $\mathcal{T}_i$, and extend this input-output framework to the original nonlinear dynamics.

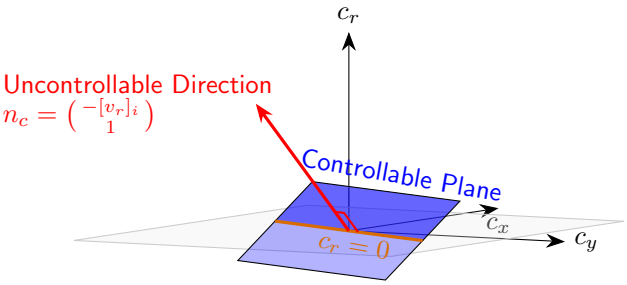


Fig. 5. Input-to-RBM map. Reachable RBMs form the blue "controllable plane," orthogonal to the uncontrollable direction n_c. Its intersection with the translation subspace ($c_r = 0$, horizontal plane) is the 1D set of perfect-recovery outcomes.

REFERENCES

Asimow, L. and Roth, B. (1978). The rigidity of graphs. *Transactions of the American Mathematical Society*, 245, 279–289.

Asimow, L. and Roth, B. (1979). The rigidity of graphs, II. *Journal of Mathematical Analysis and Applications*, 68(1), 171–190.

Babazadeh, R. and Selmic, R. (2019). Distance-based multiagent formation control with energy constraints using sdre. *IEEE Transactions on Aerospace and Electronic Systems*, 56(1), 41–56.

Chung, S.J., Paranjape, A.A., Dames, P., Shen, S., and Kumar, V. (2018). A survey on aerial swarm robotics. *IEEE Transactions on robotics*, 34(4), 837–855.

Connelly, R. (2005). Generic global rigidity. *Discrete & Computational Geometry*, 33, 549–563.

Krick, L., Broucke, M.E., and Francis, B.A. (2009). Stabilisation of infinitesimally rigid formations of multi-robot networks. *International Journal of Control*, 82(3), 423–439.

Laman, G. (1970). On graphs and rigidity of plane skeletal structures. *Journal of Engineering mathematics*, 4(4), 331–340.

Oh, K.K. and Ahn, H.S. (2011). Formation control of mobile agents based on inter-agent distance dynamics. *Automatica*, 47(10), 2306–2312.

Oh, K.K., Park, M.C., and Ahn, H.S. (2015). A survey of multi-agent formation control. *Automatica*, 53, 424–440.

Rahmani, A., Ji, M., Mesbahi, M., and Egerstedt, M. (2009). Controllability of multi-agent systems from a graph-theoretic perspective. *SIAM Journal on Control and Optimization*, 48(1), 162–186.

Skogestad, S. and Postlethwaite, I. (2005). *Multivariable Feedback Control: Analysis and Design*. John Wiley & Sons, Inc., Hoboken, NJ, USA.

Sun, Z., Mou, S., Anderson, B.D., and Cao, M. (2016). Exponential stability for formation control systems with generalized controllers: A unified approach. *Systems Control Letters*, 93, 50–57.

Tanner, H.G. (2004). On the controllability of nearest neighbor interconnections. In *43rd IEEE Conference on Decision and Control*, volume 3, 2467–2472. IEEE.